Task 5: Exploratory Data Analysis (EDA)

Titanic Passenger Dataset | Data Analyst Internship

Dataset: 891 passenger records and 12 variables. Overall survival rate: **38.4%**. Median age: **28.0 years**. Fare skewness: **4.79**.

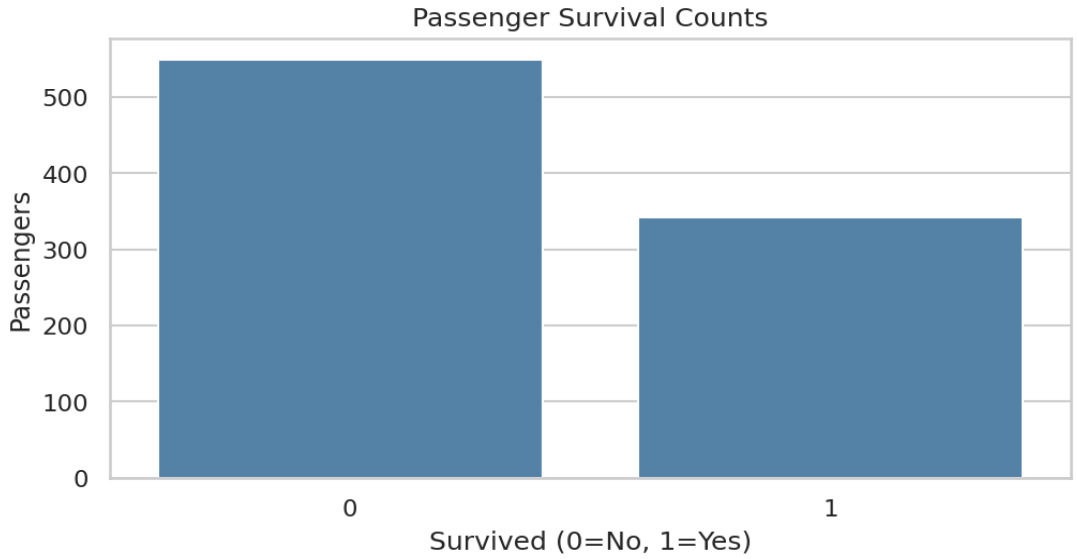
1. Data quality and statistical overview

Measure	Value
Rows	891
Columns	12
Survival rate	38.4%
Age missing	177
Cabin missing	687
Embarked missing	2
Duplicate rows	0

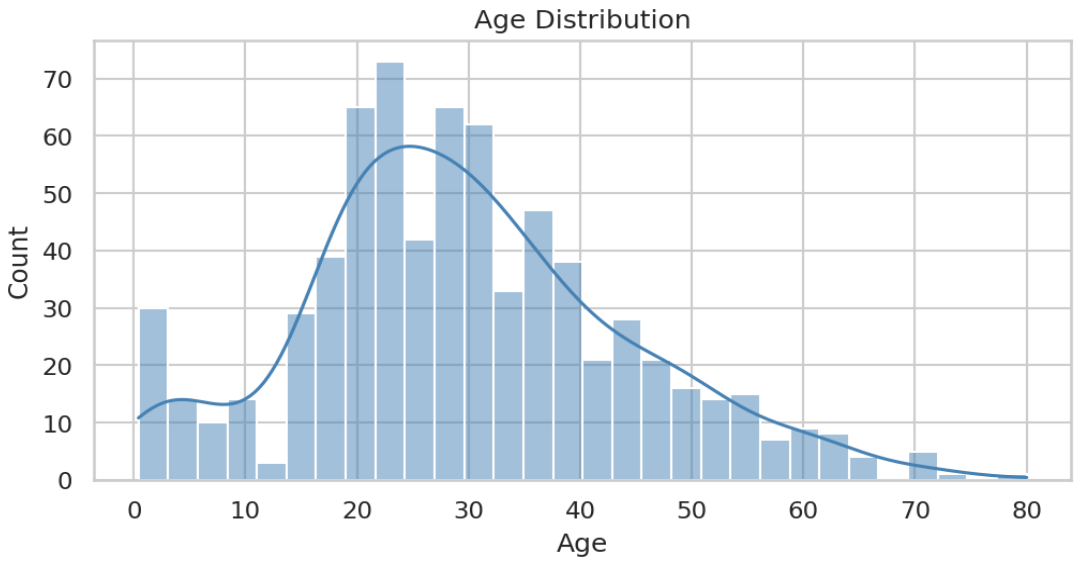
Cabin has extensive missingness, Age has some missing values, and Embarked has only a small number. Cabin should not be used naively without a missing-data strategy.

2. Visualizations and observations

1. Survival counts show that non-survivors outnumber survivors.



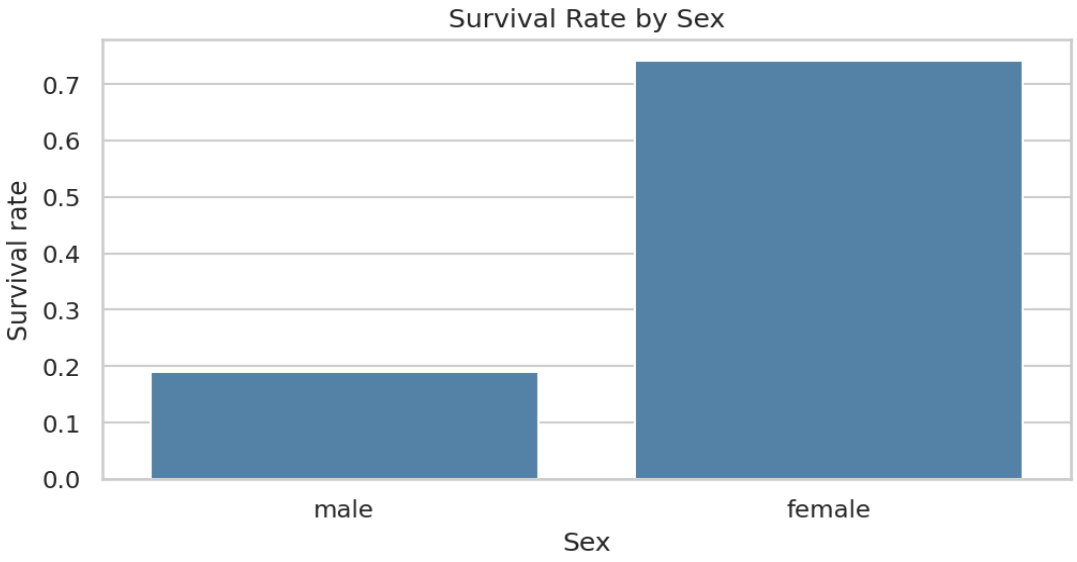
2. Age is concentrated among adults, with fewer children and older passengers.



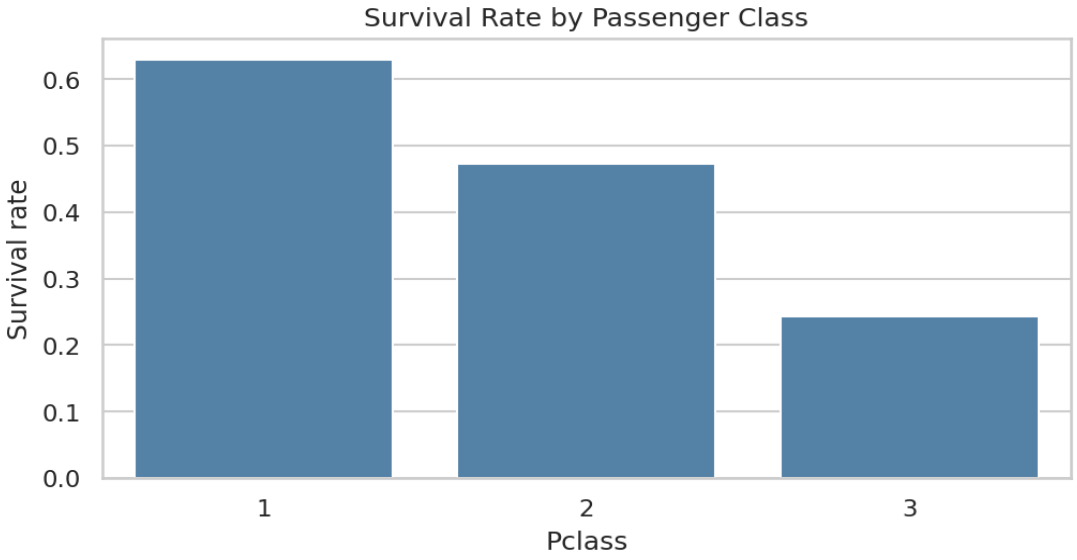
3. Fare distributions differ by class and include high-value outliers; fare is right-skewed.



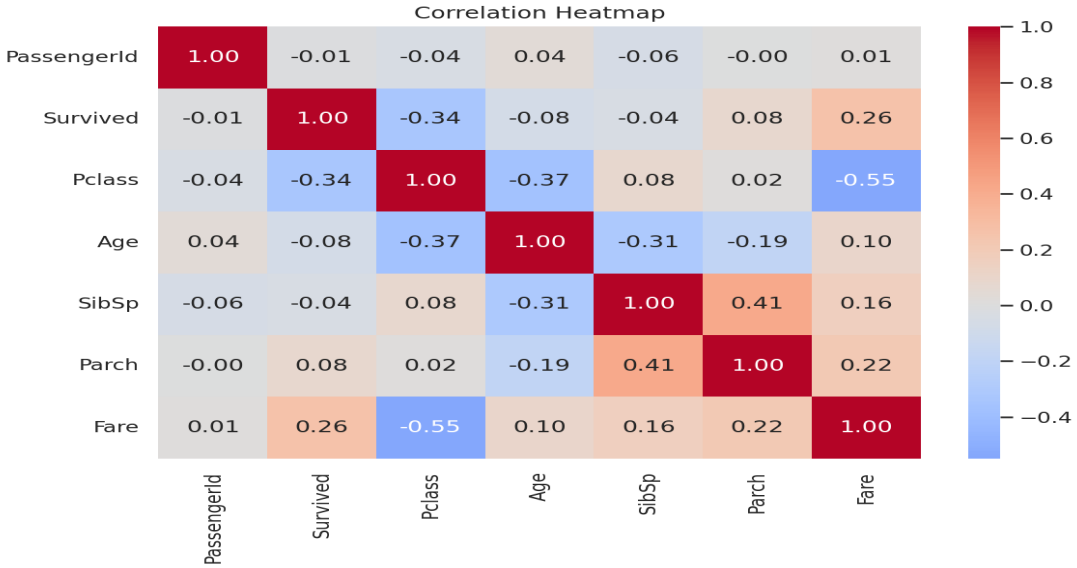
4. Female survival rate: 74.2%; male survival rate: 18.9%.



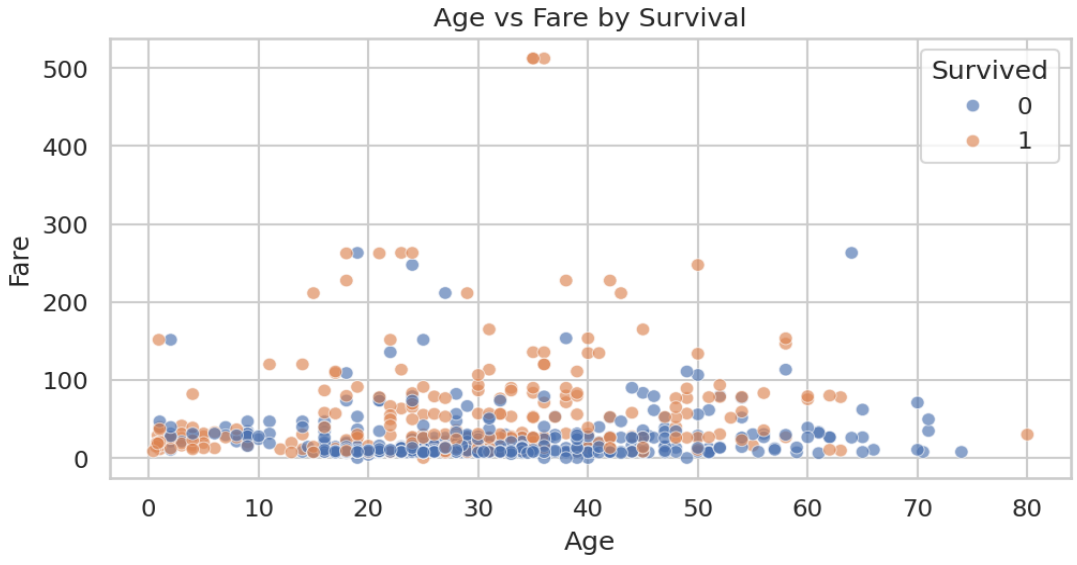
5. Survival rates: first class 63.0%, second class 47.3%, third class 24.2%.



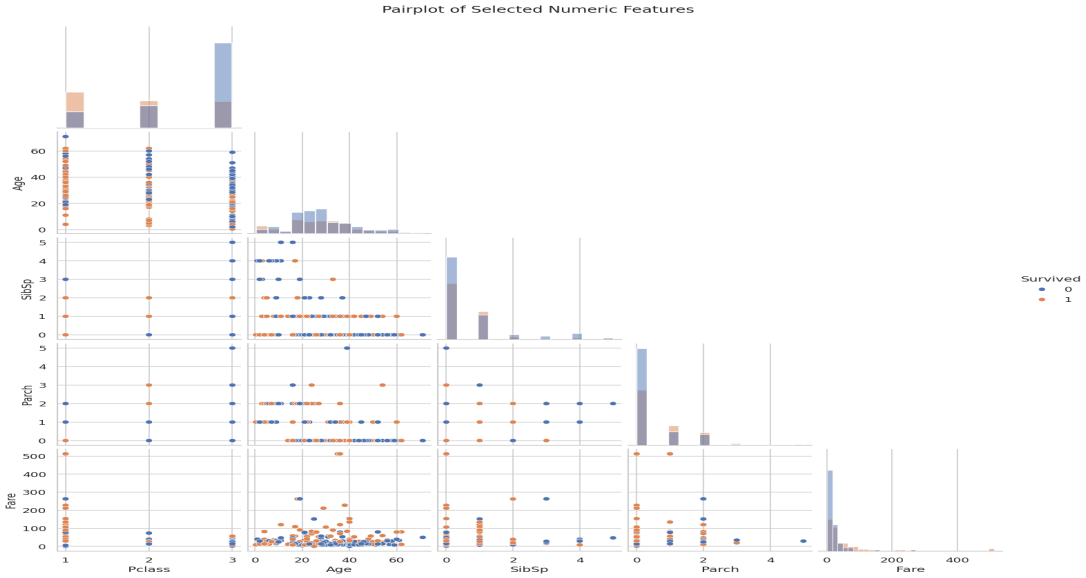
6. Heatmap summarizes pairwise correlations among numeric variables; correlation does not prove causality.



7. Age and fare do not show a simple linear relationship; fare values are widely dispersed.



8. Pairplot compares distributions and pairwise relationships across selected numeric variables.


Pairplot of Selected Numeric Features
Survived
0
1
Pclass
Age
SibSp
Parch
Fare

3. Key findings

• Overall survival rate: 38.4%.
• Women had a much higher observed survival rate than men.
• First-class passengers had the highest observed survival rate, followed by second and third class.
• Fare is strongly right-skewed; median and IQR are useful alongside mean and standard deviation.
• Cabin and Age contain missing values and require deliberate treatment.

4. Recommendations and limitations

For further analysis, explore survival by sex and class jointly, derive family-size features from SibSp and Parch, and use VIF if preparing a regression model. This report describes associations in the dataset; it does not establish causation or predict individual outcomes.

Dataset source: Data Science Dojo Titanic CSV – https://github.com/datasciencedojo/datasets/blob/master/titanic.csv